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① A) Create the following relation for the student: student (regno: string, name: string, class: string, bdate: date, marks1: int, marks2: int, marks3: int)

(i) Enter at least 5 tuples of the above relation

(ii) Demonstrate the usage of following clauses for the above relation

a) sum  c. count  e) between
b) Avg  d. like  f) max & min

(i) Enter atleast five tuples of the above relation

```
Create table student (regno int, name varchar(8),
  bdate date, marks1 int, marks2 int, marks3 int);

insert into student values (1, 'AAA', '10-AUG-2000',
   99,89,99);

insert into student values (2, 'AA', '10-APRIL-2000',
   99,89,99);

insert into student values (2, 'AAB', '10-NOV-2000'
   99,89,99);

insert into student values (3, 'AFB', '10-JULY-2000'
   99,89,98);

insert into student values (4, 'AFB', '10-JAN-2000',
   99,82,98);
```

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```
insert into student values (5, 'AQFB', '1-JAN-2000',
   94,82,98);

select * from student;
```

| REGNO | NAME | BDATE | MARKS1 | MARKS 2 | MARKS3 |
|---|---|---|---|---|---|
| 1 | aaa | 10-AUG-00 | 99 | 89 | 99 |
| 2 | aa | 10-APR-00 | 99 | 89 | 99 |
| 2 | aab | 10-NOV-00 | 99 | 89 | 99 |
| 3 | afb | 10-JUL-00 | 99 | 89 | 98 |
| 4 | afb | 10-JAN-00 | 99 | 82 | 98 |
| 5 | aqfb | 01-JAN-00 | 94 | 82 | 98 |

(ii) Demonstrate the usage of following clauses for the above relation

(a) WHERE

```
select bdate from student where bdate between
'10-JAN-2000' AND '10-DEC-2000';
```

BDATE
10-AUG-00
10-APR-00
10-NOV-00
10-JUL-00
10-JAN-00

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b) Order by

select bdate from student order by bdate Asc;

B DATE
01-JAN-00
10-JAN-00
10-APR-00
10-JUL-00
10-AUG-00
10-NOV-00

c) Group by

ex 1: Select name from student GroupBy Name;

Name
aaa
aa
a|b
aq|b
aab

Ex 2: Select count(name) from student where name in ('AAA', 'AA', 'AAB') orderBy marks2;

Count(Name)
3



d. Having
select max(marks1) "Highest marks" from student group by name having name = 'AAA';

Highest Marks
99

(iii) Demonstrate the usage of following clauses for the above relation

(a) Sum
select sum(marks1) "Total Marks" from student;

Total Marks
589.666667

(b) avg
select avg(marks1) "AVERAGE MARKS" From student;

Total Marks
98.1666667

(c) Count
select count(Marks1) "Total Count" From student;

Total Count
6

d) Like

select marks1, marks2, marks3 from student
where name like '%B' order by name;

| MARKS1 | MARKS2 | MARKS3 |
|---|---|---|
| 99 | 89 | 99 |
| 99 | 89 | 98 |
| 99 | 82 | 98 |
| 94 | 82 | 98 |

e) Between

select count(marks2) from student group by name
having count(RegNO) Between 1 AND 10;

MARKS2
------
1
1
2
1
1

(f) MAX & MIN

select MAX(Marks2) from student;

MAX(Marks2)
--------
89

select MIN(Marks2) from student;

MIN(MARKS2)
----------
82



(iv) Demonstrate the rollback and commit command for the above relation

ROLLBACK;
Roll back complete

COMMIT;
commit complete

1 B) Consider the following database that maintain information about employees & Departments
Employee (empid : int, ename : string, age : int, salary : int, deptno : int)
Department (deptno : int, dname : string, manager id : int)
(i) Create the above tables by properly specifying the primary key & foreign keys
(ii) Enter at least 5 tuples for each relation
(iii) Display emp-id, emp name whose salary lies between 5000 and 10,000
(iv) List empname & salary for all the employee working for BCA Dept.
(v) Display empname & deptname for all the manager.

(i) Create the above tables by properly specifying the primary keys & foreign keys

create table employee (empid int, ename varchar(10), age int, salary int, deptno int primary key);

create table dept (deptno references employee, dname var char(10), mid varchar(10));

(ii) Enter at least 5 tuples for each relation
insert into employee values (1, 'abc', 20, 10000, 101);

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insert into employee values (2, 'abd', 21, 10000, 1
insert into employee values (3, 'abc', 20, 10300, 1
insert into employee values (4, 'abx', 21, 10300, 104
insert into employee values (5, 'abde', 21, 7000, 1

select * from employee;

| EMPID | ENAME | AGE | SALARY | DEPTNO |
|---|---|---|---|---|
| 1 | abc | 20 | 10000 | 101 |
| 2 | abd | 21 | 10000 | 102 |
| 3 | abe | 20 | 10300 | 103 |
| 4 | aby | 21 | 10300 | 104 |
| 5 | abde | 21 | 7000 | 105 |

insert into dept values (101, 'bca', 80);
insert into dept values (103, 'bcom', 81);
insert into dept values (102, 'bcom', 83);
insert into dept values (104, 'bcom', 84);
insert into dept values (104, 'bcom', 85);

select * from dept;

| DEPT NO | DNAME | MID |
|---|---|---|
| 101 | bca | 80 |
| 103 | bcom | 81 |
| 102 | bcom | 83 |
| 104 | bcom | 84 |
| 104 | bcom | 85 |

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(iii) display emp-id & emp name whose salary lies between 5000 and 10000

| ENPID | ENAME |
|---|---|
| 1 | abc |
| 2 | abd |
| 5 | abde |

(iv) List empname, salary for all the employee working in the BCA dept

select salary, ename from emp, dpt where dname = 'bca' and emp.dptno = dpt.dptno;

| ENAME | Salary |
|---|---|
| abc | 10000 |

(v) Display empname & dept name & deptname for all the manager

Select ename, dname, mid from emp, dpt where emp.dptno = dpt.dptno

| Ename | Dname | mid |
|---|---|---|
| abc | bca | 80 |
| abd | bcom | 81 |
| abe | bcom | 83 |
| aby | bcom | 84 |
| abde | bcom | 85 |



02. Consider the following schema for order Database:
SALESMAN (Salesman-id, Name, City, commission)
CUSTOMER (Customer-id, Cust-Name, City, Grade, Salesman-id)
ORDERS (Ord-No, Purchase-Amt, Ord-Date, #customer-id, salesman-id)
Write SQL queries to
(i) Count the customers with grades above Belgavi's average
(ii) Find the name and numbers of all salesman who had more than one customer.
(iii) List all salesman and indicate those who have and dont have customers in their cities (Use UNION operation).
(iv) Create a view that finds the salesman who has the customer with the highest order of a day
(v) Demonstrate the DELETE operation by removing salesman with id # all his orders must also be deleted.

create table salesman(sid int primary key, sname varchar(10), city varchar(10), comm int);

Create table cust(cid int primary key, cname varchar(10), grade int(10), sid references salesman city varchar(10));

create table ord(ono int, pamt int, odate date, cid references cust, sid references salesman);

insert into salesman values (1, 'prakash', 'Bangalore', 1000);

insert into salesman values (2, 'pramod', 'Hubli', 2000);

insert into salesman values (3, 'leela', 'Hubli', 3000);
insert into salesman values (4, 'reshma', 'Mangalore', 5000);
insert into salesman values ( , 'reshma', 'Raichur', 5000);

Select * from salesman;

| SID | Sname | City | COMM |
|---|---|---|---|
| 1. | Prakash | Bangalore | 1000 |
| 2. | Pramod | Hubli | 2000 |
| 3 | leela | Hubli | 3000 |
| 4. | reshma | Mangalore | 5000 |
| 5. | reshma | Raichur | 5000 |

insert into cust values (50, 'raju', 100, 1, 'Dharwad');
insert into cust values (51, 'rahul', 90, 2, 'mysuru');
insert into cust values (52, 'raksha', 80, 3, 'Belgavi');
insert into cust values (53, 'ranjan', 90, 4, 'Belgavi');
insert into cust values (54, 'reema', 80, 5, 'Bangalore');

select * from cust;

| CID | CNAME | GRADE | SID | CITY |
|---|---|---|---|---|
| 50 | raju | 100 | 1 | Dharwad |
| 51 | rahul | 90 | 2 | mysur |
| 52 | raksha | 80 | 3 | Belgavi |
| 53 | ranjan | 90 | 4 | Belgavi |
| 54 | reema | 80 | 5 | Bangalore |



select * from cust;

| CID | CNAME | GRADE | SID | CITY |
|---|---|---|---|---|
| 50 | raju | 100 | 1 | Dharwad |
| 51 | rahul | 90 | 2 | mysur |
| 52 | raksha | 80 | 3 | Belagavi |
| 53 | ranjan | 90 | 4 | Belgavi |
| 54 | reema | 80 | 5 | Bengalore |

insert into ord values (100, 10000, '22-July-2022', 50, 1);

insert into ord values (101, 10000, '22-june-2022', 51, 2);

insert into ord values (102, 10001, '2-Jan-2022', 52, 3);

insert into ord values (103, 1000, '20-jan-2022', 53, 4);

insert into ord values (104, 1000, '20-feb-2022', 54, 5);

select * from ord;

| ONO | PAMT | ODATE | CID | SID |
|---|---|---|---|---|
| 100 | 10000 | 22-JUL-22 | 50 | 1 |
| 101 | 10000 | 22-JUN-22 | 51 | 2 |
| 102 | 10001 | 02-JAN-22 | 52 | 3 |
| 103 | 1000 | 20-JAN-22 | 53 | 4 |
| 104 | 1000 | 20-FEB-22 | 54 | 5 |

(i) Count the customers with grades above Belgavi's average

select count(*), grade from cust group by grade having grade > (select avg(grade) from cust where city = 'Belgavi');

| count(*) | Grade |
|---|---|
| 1 | 100 |
| 2 | 90 |

(ii) Find the name and numbers of all salesman who had more than one customer

select salesman.sid, salesman.sname from salesman, cust where 1 < (select count(*) from cust where cust.sid = salesman.sid);

> no rows selected

(iii) List all salesman and indicate those who have and don't have customers in their cities (use UNION operation)

select s.sid, s.sname, c.cname, s.comm from salesman s, cust c where s.city = c.city union (select sid, sname, 'no', comm from salesman where not city = any(select city from cust));



| SID | SNAME | CNAME | COMM |
|---|---|---|---|
| 1 | prakash | reema | 1000 |
| 2 | pramod | no | 2000 |
| 3 | leela | no | 3000 |
| 4 | reshma | no | 5000 |
| 5 | reshma | no | 5000 |

(iv) Create a view that finds the salesman who has the customer with the highest order of a day

create view new_view as select o.odate, s.sid, s.sname from salesman s, ord o, cust c where s.sid = o.sid and s.sid = c.sid and o.pampt = (select max(pampt) from ord o where c.cid = o.cid);

>> view created

select * from new_view;

| ODATE | SID | SNAME |
|---|---|---|
| 02-JAN-22 | 3 | leela |
| 20-JAN-22 | 4 | reshma |
| 22-FEB-22 | 5 | reshma |
| 22-JUL-22 | 1 | prakash |
| 22-JUN-22 | 2 | pramod |

(v) Demonstrate the DELETE operation by removing salesman with id 4.

Delete from salesman where sid = 4;

03. Insurance Database

consider the Insurance database given below. The primary keys are underlined and the datatypes are specified

PERSON (driver-id : string, name : string, address : string)

CAR (regno : string, model : string, year : int)

ACCIDENT (report-number : int, date : date, location : string)

OWNS (driver-id : string, regno : string)

PARTICIPATED (driverid : string, regno : string, report-number : int, damage-amount : int) :

(i) Create the above tables by properly specifying the primary keys and the foreign keys.

(ii) Enter atleast five tuples for each relation

(iii) Demonstrate how you

(a) Update the damage amount for the car with a specific regno in accident with report number 312 to 25000

(b) Add a new accident to the database

(iv) Find the total number of people who owned cars that were involved in accidents in the year 2006

(v) Find the number of accidents in which cars belonging to a specific model were involved



(i) Create the above tables by properly specifying the primary keys and the foreign keys.

```
create table person
 (
 driverid varchar(5) primary key,
 name varchar(10),
 address varchar(10)
 );
create table car
 (
 regno varchar(5) primary key,
 model varchar(12),
 year int);
Create table accident
 (
 reportno int primary key,
 AccidentDate date,
 location varchar(10)
 );
Create table owns
 (
 driverid references person,
 regno references car
 );
Create table participated
 {
```

```
{driverid references person,
 regno references car,
 reportno references accident,
 damages int);
```

(ii) Enter atleast five tuples for each relation

```
insert into person values(1,'Baskar','bhatkal');
insert into person values(2,'Amar','bhatkal');
insert into person values(4,'Akbar','Byndoor');
insert into person values(5,'Antony','Kumta');
select * from person;
```

| DRIVERID | NAME | ADDRESS |
|---|---|---|
| 1 | Baskar | bhatkal |
| 2 | Amar | bhatkal |
| 3 | Amar | Honnavar |
| 4 | Akbar | Byndoor |
| 5 | Antony | Kumta |

```
insert into car values('a1','merc',2007);
insert into car values('a2','porsche',2005);
insert into car values('b2','rolls royce',2006);
insert into car values('b4','ferrari',1999);
insert into car values('c1','bentley',2000);
```



```
select * from car;
```

| REGNO | MODEL | YEAR |
|---|---|---|
| a1 | Merc | 2007 |
| a2 | Porsche | 2005 |
| b2 | rolls royce | 2006 |
| b4 | Ferrari | 1999 |
| c1 | Bentley | 2000 |

```
insert into accident values(123,'1-jan-2006',
 'Mysooru');
insert into accident values(222,'28-mar-2006',
 'Hubli');
insert into accident values(312,'2-dec-2006',
 'karwar');
insert into accident values(454,'5-Jan-2007,
 yelhanka');
insert into accident values(124,'24-Jan-2007'
 'Kumta');

select * from accident;
```

| REPORTNO | ACCIDENTDATE | LOCATION |
|---|---|---|
| 123 | 1-Jan-2006 | Mysooru |
| 222 | 28-mar-2006 | Hubli |
| 312 | 2-dec-2006 | Karwar |
| 454 | 5-Jan-2007 | yalahanka |
| 124 | 26-Jan-2007 | Kumta. |

```
insert into owns values (1, 'a1');
insert into owns values (2, 'a2');
insert into owns values ( 3, 'b2');
insert into owns values (4, 'b4');
insert into owns values (5, 'c1');

select * from owns;
```

| DRIVERID | REGNO |
|---|---|
| 1 | a1 |
| 2 | a2 |
| 3 | b2 |
| 4 | b4 |
| 5 | c1 |

```
insert into participated values (1, 'a1', 123, 500);
insert into participated values (2, 'a2', 222, 2000);
insert into participated values ( 3, 'b2', 312, 1000);
insert into participated values (4, 'b4', 454, 1500);
insert into participated values (5, 'c1', 124, 800);

select * from participated;
```

| DELIVERID | REGNO | REPORTNO | DAMAGES |
|---|---|---|---|
| 1 | a1 | 123 | 500 |
| 2 | a2 | 222 | 2000 |
| 3 | b2 | 312 | 1000 |
| 4 | b4 | 454 | 1500 |
| 5 | c1 | 124 | 800 |

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(iii) Demonstrate how you
a) update the damage amount for the car with a specific regno in accident with report number 312 to 25000

```
update participated set damages = 25000 where reportno = 312;

select * from participated;
```

| DRIVERID | REGNO | REPORTNO | DAMAGES |
|---|---|---|---|
| 1 | a1 | 123 | 500 |
| 2 | a2 | 222 | 2000 |
| 3 | b2 | 312 | 25000 |
| 4 | b4 | 454 | 1500 |
| 5 | c1 | 124 | 800 |

b. Add a new accident to the database

```
insert into accident value (125, '26-jan-2007', 'Kumta');

select * from accident;
```

| REPORTNO | ACCIDENTDATE | LOCATION |
|---|---|---|
| 123 | 1-JAN-2006 | Mysooru |
| 222 | 28-mar-2006 | Hubli |
| 312 | 2-dec-2006 | Karwar |
| 454 | 5-jan-2007 | yelhanka |
| 124 | 26-jan-2007 | Kumta |
| 125 | 26-jan-2007 | Kumta. |

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(iv) Find the total number of people who owned cars that were involved in accidents in 2006.

select count(*) "No. of people" from person, car, accident, participated where person.driverid = participated where person.driverid and accident.reportno = participated.reportno and car.regno = participated.regno and AccidentDate >= '01-jan-2006' and AccidentDate <= '31-dec-2006';

No. of people
3

(v) Find the number of accidents in which cars belonging to a specific model (For example model = Porsche) were involved.

select count(*) "no. of accident" from accident where reportno in (select reportno from participated where regno in (select regno from car where model = 'porsche'));

no. of accident
1.

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Q4. Consider the following relations for the details maintained by a book dealer
AUTHOR (Author-id: int, Name: string, City: string, Country: string)
PUBLISHER (Publisher-id, int, Name: string, city: string, Country: string)
CATEGORY (category id, int, Description: string)
CATALOG (Book-id: int, title: string, author-id: int publisher-id: int, category-id: int, Year: int, Book id: int, Quantity: int)

(i) Create the above tables by properly specifying the primary keys and the foreign keys
(ii) Enter at least five tuples for each relation
(iii) Give the details of the authors who have two or more books in the catalog and the year of publications is after 2000
(iv) Find the author of the book which has the maximum sales
(v) Demonstrate how you increase the price of books published by a specific publisher by 10%.

(i) Create the above tables by properly specifying the primary keys and the foreign keys

create table author20
(
aid int primary key,
name varchar(10),

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city varchar(10),
country varchar(10)
);

create table publisher20
(
pid int primary key,
name varchar(10),
city varchar(10),
country varchar (10)
);

create table category20
(
cid int primary key,
description varchar(10)
);

create table catalog20
(
bid int primary key,
title varchar(10),
aid references author20,
pid references publisher20,
cid references category20,
year int,
price int);

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create table odetails20
(
ono int,
bid reference catalog20,
qty int );

(ii) Enter at least five tuples for each relations

insert into author20 values(1, 'mik', 'bang', 'ind');
insert into author20 values(2, 'muj', 'bang', 'ind');
insert into author20 values(3, 'prad', 'tri', 'aus');
insert into author20 values(4, 'maj', 'anan', 'euro');

insert into publisher20 values(101, 'pearson', 'bang', 'ind');
insert into publisher20 values(102, 'tata', 'mumbai' 'aus');
insert into publisher20 values(103, 'sapna', 'che' 'euro');
insert into publisher20 values(104, 'abc', 'tri', 'ame');
insert into publisher20 values(105, 'xyz', 'anan', ind);

insert into category20 values(1001, 'computer');
insert into category20 values(1002, electronics');
insert into category20 values(1003, 'maths');
insert into category20 values(1004, 'science
insert into category20 values(1005, 'electrical');

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```
insert into catalog20 values(111,'lib1',1,101,1001,2002,500);
insert into catalog20 values(112,'lib2',2,102,1002,2000,800);
insert into catalog20 values(113,'lib3',3,103,1003,2003,
200);
insert into catalog20 values(114,'lib4',4,104,1001,2006
350);
insert into catalog20 values(115,'lib5',5,105,1004,2007,
100);
insert into catalog20 values(116,'lib6',2,103,1005,2007,
100);
insert into catalog20 values(117,'lib7',2,105,1002,2007,
450);
insert into catalog20 values(118,'lib8',1,101,1001,2002
500);

insert into odetails20 values(1,111,2);
insert into odetails20 values(2,112,3);
insert into odetails20 values(3,111,5);
insert into odetails20 values(4,113,1);
insert into odetails20 values(5,114,2);
insert into odetails20 values(6,115,21);
insert into odetails20 values(1,114,2);
insert into odetails20 values(2,113,2);

select * from author20;
```

| AID | NAME | CITY | COUNTRY |
|---|---|---|---|
| 1 | mik | bang | ind |
| 2 | muj | bang | ind |
| 3 | prad | tri | aus |
| 4 | maj | anan | ame |



| | | | |
|---|---|---|---|
| 5 | waj | anan | euro |

```
select * from publisher20;
```

| PID | NAME | CITY | COUNTRY |
|---|---|---|---|
| 101 | pearson | bang | ind |
| 102 | tata | mumbai | aus |
| 103 | sapna | che | euro |
| 104 | abc | tri | ame |
| 105 | xyz | anan | ind |

```
select * from category20;
```

| CID | DESCRIPTION |
|---|---|
| 1001 | computer |
| 1002 | electronics |
| 1003 | maths |
| 1004 | science |
| 1005 | electrical |

```
select * from catalog20;
```

| BID | TITLE | AID | PID | CID | YEAR | PRICE |
|---|---|---|---|---|---|---|
| 111 | lib1 | 1 | 101 | 1001 | 2002 | 500 |
| 112 | lib2 | 2 | 102 | 1002 | 2000 | 800 |
| 113 | lib3 | 3 | 103 | 1003 | 2003 | 200 |
| 114 | lib4 | 4 | 104 | 1001 | 2006 | 350 |
| 115 | lib5 | 5 | 105 | 1004 | 2007 | 100 |
| 116 | lib6 | 2 | 103 | 1005 | 2007 | 600 |
| 117 | lib7 | 2 | 105 | 1002 | 2007 | 450 |

select * from odetails20;

| ONO | BID | QTY |
|---|---|---|
| 1 | 111 | 2 |
| 2 | 112 | 3 |
| 3 | 113 | 1 |
| 4 | 114 | 2 |
| 5 | 115 | 1 |
| 6 | 114 | 2 |
| 1 | 113 | 2 |
| 2 | 111 | 5 |
| 3 | 112 | 1 |

(iii) Give the details of the authors who have 2 or more books in the catalog and the price of the books is greater than the average price of the books in the catalog and the year of publication is after 2000

select name, city, country from author20 where aid in(select aid from catalog20 where year >2000 and price > (select avg(price) from catalog20) group by aid having count(*) >1);

| Name | CITY | COUNTRY |
|---|---|---|
| mik | bang | ind |
| muj | bang | ind |



(iv) Find the author of the book which has maximum sales

select a.aid, name from author20 a, catalog20 c where a.aid = c.aid and c.bid = (select bid from odetails20 group by bid having sum(qty) = (select max(sum(qty)) from odetails20 group by bid));

| AID | NAME |
|---|---|
| 1 | mik |

(V) Demonstrate how you increase the price of books published by a specific publisher by 10%.

update catalog20 set price = 1.1 * price where pid in (select pid from publisher20 where name = 'abc');

1 row updated

SQL> select * from catalog20;

| BID | TITLE | AID | PID | CID | YEAR | PRICE |
|---|---|---|---|---|---|---|
| 111 | lib1 | 1 | 101 | 1001 | 2002 | 500 |
| 112 | lib2 | 2 | 102 | 1002 | 2000 | 800 |
| 113 | lib3 | 3 | 103 | 1003 | 2003 | 200 |
| 114 | lib4 | 4 | 104 | 1001 | 2006 | 385 |
| 115 | lib5 | 5 | 105 | 1004 | 2007 | 100 |
| 116 | lib6 | 2 | 103 | 1005 | 2007 | 600 |
| 117 | lib7 | 2 | 105 | 1002 | 2007 | 450 |
| 118 | lib8 | 1 | 101 | 1001 | 2002 | 500 |

5. Consider the following database of student enrollment in courses and books adopted for each courses

STUDENT (regno: string, name: string, major: string, bdate int)

COURSE (course# : int, cname : string, dept : string)

ENROLL (regno: string, course# : int, sem : int, marks int)

TEXT (book_isbn : int, booktitle : string, publisher : string, author : string)

BOOK_ADOPION (course # : int, sem : int, book_isbn : int)

(i) Create the above tables by properly specifying the primary keys and the foreign key

```
create table student (regno varchar (5) primary key, name varchar(10), major varchar (10), bdate date);
create table course (cno int primary key, cname varchar (10), dept varchar(10));
create table enroll (regno reference student, cno reference course, sem int primary key, marks int);
create table text (isbn int primary key, title varchar(15), publisher varchar(10), author varchar(10));
create table book (cno reference course, sem reference enroll, isbn reference text);
```

(ii) Enter atleast five tuples for each relation.

```
insert into student values ('cs42', 'mikhil', 'cse', ('17-dec-1986'));
insert into student values('cs48', 'mujeeb', 'cse', '02-sep-1986');
insert into student values('ec26', 'pradeep', 'cse', '16-aug-1987');
insert into student values('ee37', 'majid', 'ise', '28-may-1986');
insert into student values ('is48', 'wajid', 'ise', '28-may-1986');

insert into course values (1, 'net', 'computer');
insert into course values (2, 'j2ee', 'computer');
insert into course values (3, 'mis', 'infosci');
insert into course values (4, 'js', 'infosci');
insert into course values (5, 'oracle', 'computer');

insert into enroll values ('cs42', 1, 6, 98);
insert into enroll values ('cs48', 2, 3, 97);
insert into enroll values ('ec26', 3, 5, 50);
insert into enroll values ('is48', 4, 7, 90);
insert into enroll values ('ee37', 5, 4, 80);
```

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```
insert into text values(101, 'net as c', 'lpe', 'jahad');
insert into text values(102, 'c++', 'abc', 'mujeeb');
insert into text values(103, 'oracle', 'del', 'othman');
insert into text values(104, 'net', 'lpe', 'naushad');
insert into text values(105, 'j2ee', 'pearson', 'mikhil');

insert into book values(1,6,101);
insert into book values(2,3,102);
insert into book values(3,5,103);
insert into book values(4,7,104);
insert into book values(5,4,105);

select * from student;
```

| REGNO | NAME | MAJOR | BDATE |
|---|---|---|---|
| CS42 | mikhil | cse | 17-DEC-86 |
| CS48 | mujeeb | cse | 02-SEP-86 |
| ec26 | pradeep | cse | 16-AUG-87 |
| ee37 | majid | ise | 28-MAY-86 |
| is48 | wajid | ise | 28-MAY-86 |

```
select * from student;
```

| REGNO | NAME | MAJOR | BDATE |
|---|---|---|---|
| CS42 | mikhil | cse | 17-DEC-86 |
| CS48 | mujeeb | cse | 02-SEP-86 |
| ec26 | pradeep | cse | 16-AUG-87 |
| ee37 | majid | ise | 28-MAY-86 |
| is48 | wajid | ise | 28-MAY-86 |

Expt. No. Date



```
select * from course;
```

| CNO | CNAME | DEPT |
|---|---|---|
| 1 | .net | computer |
| 2 | j2ee | computer |
| 3 | mis | infosci |
| 4 | js | infosci |
| 5 | oracle | computer |

```
select * from enroll;
```

| REGNO | CNO | SEM | MARKS |
|---|---|---|---|
| CS42 | 1 | 6 | 98 |
| CS48 | 2 | 3 | 97 |
| ec26 | 3 | 5 | 50 |
| is48 | 4 | 7 | 90 |
| ee37 | 5 | 4 | 80 |

```
select * from text;
```

| ISBN | TITLE | PUBLISHER | AUTHOR |
|---|---|---|---|
| 102 | c++ | abc | mujeeb |
| 103 | oracle | del | othman |
| 104 | .net | lpe | naushad |
| 105 | j2ee | pearson | mikhil |

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select * from book;

| CNO | SEM | ISBN |
|---|---|---|
| 1 | 6 | 101 |
| 2 | 3 | 102 |
| 3 | 5 | 103 |
| 4 | 7 | 104 |
| 5 | 4 | 105 |

(iii) Demonstrate how you add a new text to the database and make this book be adopted by some department

insert into text values(106, 'java', 'pearson', 'avril');
insert into book values(6, 7, 106);

select * from text;

| ISBN | TITLE | PUBLISHER | AUTHOR |
|---|---|---|---|
| 101 | let us c | lpe | fahad |
| 102 | C++ | abc | myjeeb |
| 103 | oracle | del | othman |
| 104 | .net | lpe | naushad |
| 105 | j2ee | pearson | mikhil |
| 106 | java | pearson | avril. |

Expt. No. Date


select * from book;

| CNO | SEM | ISBN |
|---|---|---|
| 1 | 6 | 101 |
| 2 | 3 | 102 |
| 3 | 5 | 103 |
| 4 | 7 | 104 |
| 5 | 4 | 105 |
| 6 | 7 | 106 |

(iv) Produce a list of text books (include course #, book_isbn, book-title) in the alphabetical order for courses offered by the cs department that use more 2 books

select course.cno, book.isbn, text.title from course, book, text where course, eno = book, eno and text. isbn = book. isbn and dept = 'compute' and course. eno in (select eno from book group by eno having count(*) > 2) order by ename;

or

select course.cno, book.isbn, text. title from course, book, text. where course.cno = book.cno and text.isbn = book.isbn and dept = 'computer' and 2 < (select count (isbn) from book where course.cno = book.cno) order by text.title;

>>> no rows selected.

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v) List any department that has all its adopted books published by specific publisher

```
select c.dept , t.publisher from course c, book b,
text t where c.cno = b.cno and t.isbn = b.isbn
and publisher = 'lpe';
```

| DEPT | PUBLISHER |
| --- | --- |
| computer | lpe. |

Seen 21/5/28.

Date

Expt. No.

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